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Key Points:

- Stratospheric polar vortex variability is sensitive to equilibrium temperatures in the lower stratosphere
- Large meridional temperature gradients in the lower stratosphere favor extreme fluxes of planetary-scale waves to the stratosphere
- Sudden stratospheric warming events are frequent when using both wavenumber-1 and wavenumber-2 topographies

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Lower-Stratospheric Control of the Frequency of Sudden Stratospheric Warming Events

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Abstract The sensitivity of stratospheric polar vortex variability to the basic-state stratospheric temperature profile is investigated by performing a parameter sweep experiment with a dry dynamical core general circulation model where the equilibrium temperature profiles in the polar lower and upper stratosphere are systematically varied. It is found that stratospheric variability is more sensitive to the temperature distribution in the lower stratosphere than in the upper stratosphere. In particular, a cold lower stratosphere favors a strong time-mean polar vortex with a large daily variability, promoting frequent sudden stratospheric warming events in the model runs forced with both wavenumber-1 and wavenumber-2 topographies. This sensitivity is explained by the control exerted by the lower-stratospheric basic state onto fluxes of planetary-scale wave activity from the troposphere to the stratosphere, confirming that the lower stratosphere can act like a valve for the upward propagation of wave activity. It is further shown that with optimal model parameters, stratospheric polar vortex climatology and variability mimicking Southern and Northern Hemisphere conditions are obtained with both wavenumber-1 and wavenumber-2 topographies.

1. Introduction

Strong polar vortices of westerly winds form in both the Southern Hemisphere (SH) and Northern Hemisphere (NH) during each hemisphere's winter season when perpetual night condition promotes a strong temperature gradient between the tropics and polar regions. However, there are large interhemispheric differences in both the climatology and temporal variability of the polar vortices. Notably, the SH polar vortex is much stronger in the austral winter (June-July-August) than the NH vortex in the boreal winter (December-January-February). The former, however, exhibits weaker intraseasonal variability in comparison to the latter. The largest variability of the SH polar vortex occurs in spring, known as stratospheric final warming, whereas the strongest variability of the NH polar vortex is observed in the middle of winter. The most intense events of NH stratospheric variability, the sudden stratospheric warming (SSW) events, which occur approximately 6 times per decade, influence surface weather (e.g., Baldwin, 2001; Kidston et al., 2015; Martineau & Son, 2015; Mitchell et al., 2013).

These interhemispheric differences are largely attributed to differences in planetary-scale wave generation and propagation (Plumb, 1989, 2010; Randel, 1988). Topography and land-sea temperature contrasts are larger in the NH, generating planetary-scale waves of larger amplitudes. The stronger wave activity is accompanied by stronger wave drag in the NH stratosphere, which prevents the zonal wind from becoming too strong which would prevent planetary-scale wave propagation from the troposphere to the stratosphere (Charney & Drazin, 1961). On the other hand, the drag from the weaker planetary-scale waves of the SH is insufficient to prevent the strengthening of the polar vortex above the range of zonal wind speed where planetary-scale wave propagation is allowed. The strong stratospheric vortex in the SH midwinter therefore suppresses the vertical propagation of planetary-scale waves, resulting in a midwinter minimum of planetary-scale wave activity (Plumb, 1989).

Driven by the need to better understand the wave-mean flow interactions regulating stratospheric polar vortex variability and its coupling to the troposphere, many studies have used simple atmospheric general circulation models (GCMs), typically based on a dynamical core GCM, with various configurations.

In these modeling studies, temperatures are relaxed to time-independent zonally symmetric temperature profiles, mimicking radiative-convective equilibrium temperature profiles, at prescribed time scales (e.g., Gerber & Polvani, 2009; Held & Suarez, 1994; Jucker et al., 2013; Kushner & Polvani, 2004; Polvani & Kushner, 2002; Taguchi et al., 2001). Taguchi et al. (2001) integrated a spectral primitive equation model with Newtonian relaxation of temperatures to an equilibrium temperature profile based on an idealized wind profile. Their parameter sweep experiments, with wavenumber-1 and wavenumber-2 topographies of various amplitudes, showed that the temporal variability of the stratospheric polar vortex tends to increase with increasing topography amplitude. The polar vortex regime that leads to the occurrence of SSW events was also obtained for a range of wavenumber-1 topography. However, no SSW events were found in the model simulations with wavenumber-2 topography.

Polvani and Kushner (2002) have investigated the impact of stratospheric thermal relaxation on the tropospheric circulation in a similar model. The polar stratospheric temperatures were relaxed to an equilibrium temperature profile that is specified with a constant lapse rate in the high latitudes and the U.S. Standard Atmosphere in the lower latitudes. Cooling the upper-stratospheric high latitudes in their study resulted in a stronger stratospheric vortex and also pulled the tropospheric jet and the associated surface wind maximum poleward. Kushner and Polvani (2004) extended this work by investigating the impact of stratospheric cooling, prescribed in the same manner, on wave-mean flow interactions. They observed a reduction of stratospheric planetary-scale wave drag when the vortex is strong and cold with enhanced deflection of wave activity toward low latitudes. The axisymmetric model runs with prescribed wave drag further suggested that full eddy feedbacks are necessary to produce a response of the tropospheric jet.

Gerber and Polvani (2009) updated Kushner and Polvani (2004) by including topographically forced stationary waves as in Taguchi et al. (2001). It was found that NH-like polar vortex variability is obtained for a narrow range of wavenumber-2 topography. However, no realistic vortex variability was observed when using wavenumber-1 topography since the resulting planetary-scale waves completely eroded the stratospheric polar vortex and prevented it from ever recovering. Sheshadri et al. (2015) made further adjustments to the model setup of Gerber and Polvani (2009) by imposing a meridional temperature gradient of radiative equilibrium temperatures in the lower stratosphere which improved the realism of zonal-mean zonal wind in the lower stratosphere. This modification may have helped to simulate SSW events with wavenumber-1 topography, in addition to the introduction of a seasonal cycle in the radiative equilibrium temperature, which reinitializes the vortex every winter. However, SSW frequency remained very small in comparison to observations when driven by wavenumber-1 topography.

Jucker et al. (2013) recently improved the realism of the time-mean stratospheric circulation in a primitive equation model by relaxing stratospheric temperatures to an equilibrium temperature profile that is derived from radiative transfer calculations. It was also found, using an analytical version of their temperature profile, that the frequency of SSWs is sensitive to the meridional structure of damping rates (Jucker et al., 2014). The model integrations with lower damping rates and warmer upper-stratospheric temperatures in the high latitudes yielded a higher frequency of SSW events. Their analytically described equilibrium temperature profile was shown in a subsequent study to produce SSW events with both wavenumber-1 and wavenumber-2 topographies (Jucker, 2016). Hitchcock and Shepherd (2013) also reported a sensitivity of the stratospheric circulation to lower-stratospheric damping time scales in a similar model. The longer time scales allowed circulation anomalies associated to SSW events to persist for a longer time in the lower stratosphere.

Other studies have also suggested that the lower stratosphere is important for the dynamical coupling between the stratosphere and troposphere since it can act as a valve, preventing or allowing the vertical propagation of planetary-scale waves (Chen & Robinson, 1992; Scott & Polvani, 2006). However, none of the aforementioned modeling studies have seriously addressed the sensitivity of mechanistic models to the lower-stratospheric radiative equilibrium temperature configuration. As mentioned earlier, Hitchcock and Shepherd (2013) did evaluate the sensitivity to radiative damping time scale in the lower stratosphere but did not modify the relaxation temperature profile in the same region. Although Jucker et al. (2014) did not explicitly alter lower-stratospheric temperatures in their model experiments, they found an enhanced SSW frequency in the model runs with a warmer lower stratosphere. Whether such relationship holds when the lower-stratospheric temperature profile is altered independently from the upper stratosphere remains

unknown. It is also important to note that despite the radiative transfer calculations of Jucker et al. (2013) that revealed a cold equilibrium temperature in the polar lower stratosphere, it was not included in their idealized profile (Jucker et al., 2014).

The major goal of this work is to evaluate and contrast the relative roles of the basic-state temperature in the polar lower and upper stratosphere in regulating the temporal variability and time-mean state of the stratospheric polar vortex in simulations using a primitive equation model. Our approach consists of carrying a parameter sweep experiment where the lower-stratospheric radiative equilibrium temperature profile is altered independently from the upper-stratospheric temperature profile. The equilibrium temperatures are varied in such a way as to cover a range of stratospheric vortex speeds and variabilities that range from SH- to NH-like conditions in observations. From this experiment, lower-stratospheric equilibrium temperature emerges as a key model parameter to control stratospheric vortex variability, with optimal settings allowing frequent SSW events with both wavenumber-1 and wavenumber-2 topographic forcings.

The design of the parameter sweep experiment is first described in section 2. Section 3.1 describes the time-mean circulation and daily variability of the stratospheric polar vortex in some selected simulations and evaluates the relative roles of upper- versus lower-stratospheric conditions to control polar vortex variability and the occurrence of SSW events. It is followed by section 3.2 which documents the dynamics of the lower-stratospheric control using Eliassen-Palm (EP) flux diagnostics. Section 3.3 considers the effect of topography on stratospheric variability. The sensitivity of vertical coupling to model parameters is then examined in section 3.4. Finally, section 3.5 identifies and evaluates the model runs that best reproduce stratospheric variability in observations. Discussion and conclusions are presented in sections 4 and 5.

2. Methodology

2.1. Observational Data

When model runs are compared to observations, ERA-Interim (Dee et al., 2011) reanalysis data are used from 1979 to 2013. All variables are available at a 1.5° × 1.5 ° horizontal resolution on 37 pressure levels from 1,000 to 1 hPa. Although not shown, the results shown here are not sensitive to the choice of data set (e.g., Martineau et al., 2016).

2.2. Model Settings

The model used in this study is a spectral atmospheric dynamical core developed by the Geophysical Fluid Dynamics Laboratory. This model integrates the primitive equations and relaxes the temperature field toward a prescribed temperature (T_{eq}) profile that mimics radiative-convective equilibrium in the troposphere and radiative equilibrium in the stratosphere. The model is dissipated by Rayleigh damping in the planetary boundary layer and hyperdiffusion in the free atmosphere. No other physical processes are considered. All simulations are made at T42 resolution with 40 unequally spaced levels, of which half of the levels are located above 100 hPa. The model top is located at 0.01 hPa, and eddy winds are gradually damped in a sponge layer above 0.5 hPa (Chen & Zurita-Gotor, 2008).

The linear temperature relaxation takes the form

$$\frac{\partial T}{\partial t} = \ldots - \frac{T - Teq}{\tau} \tag{1}$$

where T is actual temperature, T_{eq} is the radiative equilibrium temperature, and τ is the relaxation time scale. In the troposphere, T_{eq} is almost identical to the one used in Held and Suarez (1994) except for a weak asymmetry between the two hemispheres that is introduced to mimic solstice conditions:

$$T_{\mathrm{eq,trop}} = \max\left\{200, \left[315 - 60\sin^2\phi - 10\sin\phi - 10\log\frac{p}{p_0}\cos^2\phi\right]\left(\frac{p}{p_0}\right)^k\right\} \tag{2}$$

In the stratosphere, an idealized temperature profile is designed to qualitatively reproduce the relaxation temperature profile of Jucker et al. (2013) which produces a realistic circulation in the stratosphere. Although Jucker et al. (2014) did provide an analytic profile based on polynomial functions, a mathematically simpler profile that may be helpful in sensitivity experiments is introduced in this study. Specifically, $T_{\mathrm{eq,strat}}$ is set to a function of the reference temperature (T_{ref}) and perturbed temperature

$$T_{\mathrm{eq,strat}}(\phi, p) = T_{\mathrm{ref}} + W_{\mathrm{S}}(\phi)T_{\mathrm{SP}}(p) + W_{\mathrm{W}}(\phi)[T_{\mathrm{WP}}^{\mathrm{lowstrat}}(p) + T_{\mathrm{WP}}^{\mathrm{highstrat}}(p)] \tag{3}$$

where subscripts SP and WP denote summer pole and winter pole, respectively.

In equation (3), the SP vertical profile is given as

$$T_{\mathrm{SP}}(p) = \begin{cases} 50 \sin\left(\pi \frac{\log(p/0.01)}{\log(100/0.01)}\right) & 1\ \mathrm{hPa} < p < 100\ \mathrm{hPa} \\ 0 & p < 1\ \mathrm{hPa} \end{cases} \tag{4}$$

T_{SP} represents a warm stratosphere in the summer hemisphere whose temperature increases from 100 hPa to 1 hPa. Above 1 hPa, T_{SP} is equal to 0; thus, $T_{\mathrm{eq,strat}}$ is equivalent to the reference temperature, T_{ref}, of 200 K as in Held and Suarez (1994).

At the winter pole, the lower-stratospheric profile is given as

$$T_{\mathrm{WP}}^{\mathrm{lowstrat}}(p) = \begin{cases} \Delta T_{\mathrm{low}} \sin^2\left(\pi \frac{\log(p/3)}{\log(300/3)}\right) & 3\ \mathrm{hPa} < p < 300\ \mathrm{hPa} \\ 0 & p < 3\ \mathrm{hPa} \end{cases} \tag{5}$$

and the upper-stratospheric profile is given as

$$T_{\mathrm{WP}}^{\mathrm{highstrat}}(p) = \begin{cases} \Delta T_{\mathrm{high}} \sin^{1/2}\left(\pi \frac{\log(p/0.03)}{\log(30/0.03)}\right) & 1\ \mathrm{hPa} < p < 30\ \mathrm{hPa} \\ 0 & p < 1\ \mathrm{hPa} \end{cases} \tag{6}$$

Unlike T_{SP}, $T_{\mathrm{WP}}^{\mathrm{lowstrat}}$ has a parameter, ΔT_{low}, that is used to set the relaxation temperature in the lower stratosphere. $T_{\mathrm{WP}}^{\mathrm{lowstrat}}$ starts at 0 K at 300 hPa, increases to ΔT_{low} at 30 hPa, and then decreases back to 0 K at 3 hPa. We note that this lower-stratospheric profile, which is applied from 300 hPa and above, differs from the previous model configurations which often introduce stratospheric temperature gradients starting above 100 hPa (Gerber & Polvani, 2009; Polvani & Kushner, 2002) or 200 hPa (Sheshadri et al., 2015). Upper-stratospheric relaxation temperatures are adjusted by altering $T_{\mathrm{WP}}^{\mathrm{highstrat}}$ which is designed to increase gradually from 0 K at 30 hPa toward ΔT_{high} at 1 hPa. Above 1 hPa, $T_{\mathrm{WP}}^{\mathrm{lowstrat}}$ and $T_{\mathrm{WP}}^{\mathrm{highstrat}}$ are both equal to 0 K and thus temperatures at mesospheric heights are relaxed to the reference temperature (T_{ref}) of 200 K.

The relaxation temperature profiles described so far express the vertical dependence of stratospheric relaxation temperatures at the winter and summer poles, without a description of the meridional transition between these two profiles. To prescribe meridional gradients of relaxation temperatures, meridional weighting functions (W_{S} and W_{W}) are introduced in equation (3). W_{S} and W_{W} are defined as below

$$W_{\mathrm{W}} = \begin{cases} 0 & -90 < \phi < 45 \\ \frac{\phi - 45}{65 - 45} & 45 < \phi < 65 \\ 1 & 65 < \phi < 90 \end{cases} \tag{7}$$

and

$$W_{\mathrm{S}} = \begin{cases} 1 - 1/2\left(\frac{\phi + 90}{45 + 90}\right) & -90 < \phi < 45 \\ 1 - 1/2 - 1/2\left(\frac{\phi - 45}{65 - 45}\right) & 45 < \phi < 65 \\ 0 & 65 < \phi < 90 \end{cases} \tag{8}$$

From 65°N to 90°N, W_{W} is equal to 1 thus $T_{\mathrm{eq,strat}}$ is identical to $T_{\mathrm{ref}} + T_{\mathrm{WP}}$. From 45°N to 65°N, the weight W_{W} applied to T_{WP} decreases linearly to 0, while the weight W_{S} applied to T_{SP} increases, resulting in a pronounced meridional $T_{\mathrm{eq,strat}}$ gradient in midlatitudes. $T_{\mathrm{eq,strat}}$ does not, however, reach $T_{\mathrm{ref}} + T_{\mathrm{SP}}$ immediately at 45°N. Instead, 50% of W_{S} is linearly decreased from 90°S to 45°N. Such gradual decrease of relaxation temperatures from the summer hemisphere pole to the winter hemisphere midlatitudes results in a more realistic stratospheric circulation in the summer hemisphere (not shown).

In this study, a parameter sweep experiment is performed by varying ΔT_{low} and ΔT_{high} independently in order to identify the region that has the greatest influence on stratospheric polar vortex variability. Specifically, ΔT_{low} is varied from −10 K to −30 K in 5 K increments and ΔT_{high} is varied from 0 K to 40 K in 20 K increments. All combinations are tested for a total of 15 model simulations. Some sample T_{eq} profiles used in the parameter sweep experiment are illustrated in Figure 1 as a function of ΔT_{low} (columns) and ΔT_{high} (rows).

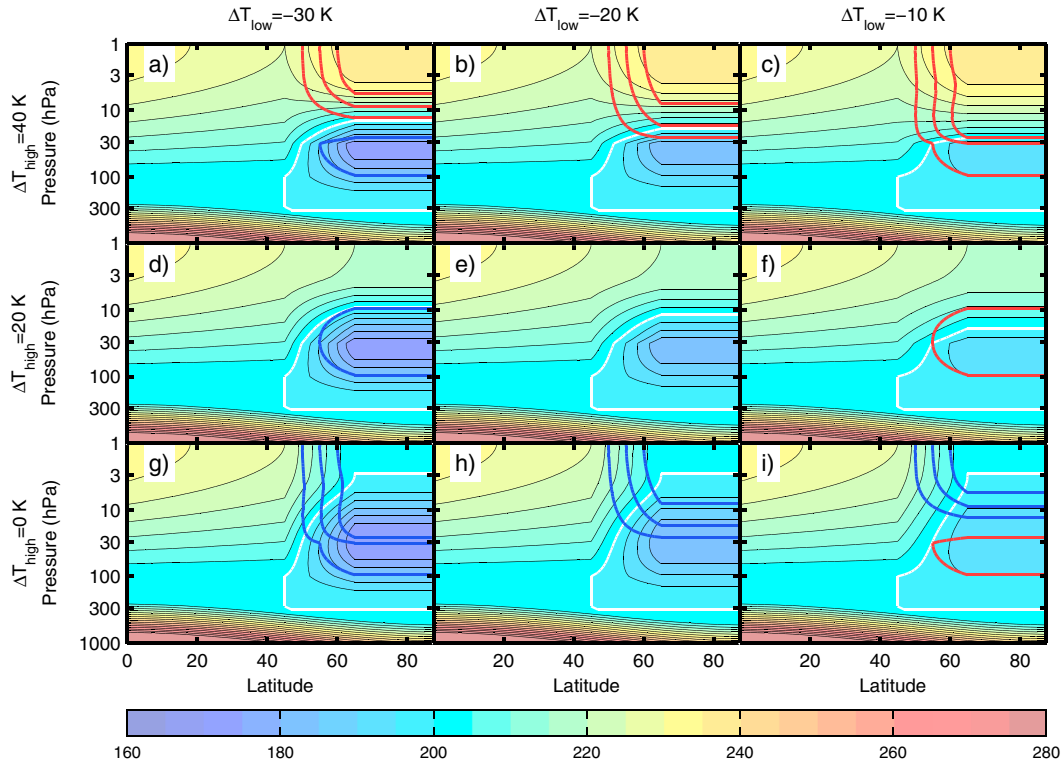

Figure 1. The equilibrium temperature profile (T_{eq}) is shown for a subset of model configurations of the parameter sweep experiment. It is shown as a function of upper-stratospheric equilibrium temperature parameter ΔT_{high} (a–c; d–f; and g–i) and lower-stratospheric equilibrium temperature parameter ΔT_{low} (a, d, g; b, e, h; and c, f, i). Color shadings indicate the equilibrium temperature with a contour interval of 5 K. The 200 K contour is shown in white. Blue and red lines are used to illustrate negative and positive T_{eq} differences with respect to the reference run (panel e; $\Delta T_{high} = 20$ K and $\Delta T_{low} = -20$ K) with a contour interval of 5 K.

The relaxation time scale, τ, also consists of tropospheric and stratospheric components. In the troposphere, τ_{trop} is identical to the one used in Held and Suarez (1994). In the stratosphere, τ_{strat} is set as below

$$\tau_{strat}(\phi, p) = \begin{cases} \max\left(40 - 35\left(1 - \frac{\log(p/10)}{\log(100/10)}\right) - 10\sin^2(\phi), 5\right) & 10\,\text{hPa} < p < 100\,\text{hPa} \\ 5 & 1\,\text{hPa} < p < 10\,\text{hPa} \\ 40 & p < 1\,\text{hPa} \end{cases} \tag{9}$$

where the unit is day. Equation (9) is essentially an idealized reproduction of Jucker et al. (2013) and Hitchcock et al. (2010). The relaxation time scale becomes progressively shorter in the stratosphere from 100 to 10 hPa (not shown) and gradually decreases by 10 days from the equator to the poles. Above 1 hPa, τ is set to 40 days as in Held and Suarez (1994).

In order to generate planetary-scale waves in the model, a simple topography is introduced following Gerber and Polvani (2009). Surface geopotential height is set as follows:

$$\Phi(\lambda, \phi) = \begin{cases} gH\sin^2\left(\frac{\phi-25}{40}\right)\cos(K\lambda) & 25 < \phi < 65 \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

where λ and ϕ refer to longitude and latitude and K and H refer to wavenumber and topography height. Two different zonal wavenumbers, that is, wavenumbers 1 and 2, are used, and various topography amplitudes from 0 km to 6 km in 1 km increments are tested. All experiments are integrated over 4,000 days, of which the first 1,000 days are excluded when evaluating the climatology and variability.

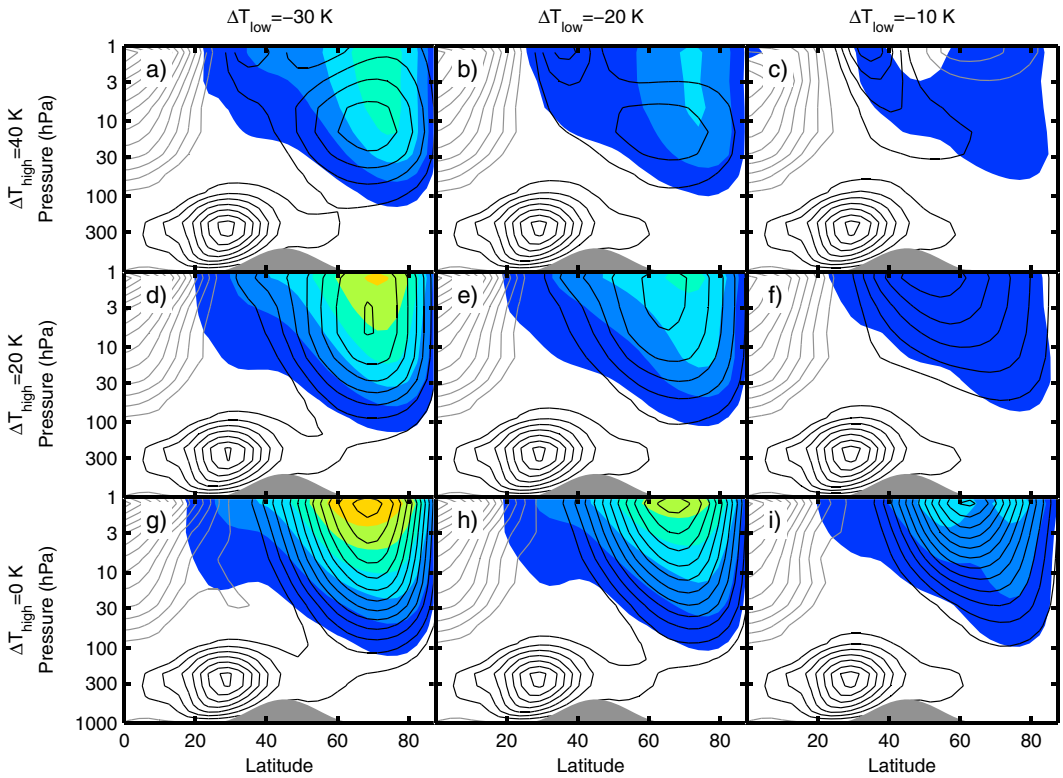

Figure 2. Climatology of zonal-mean zonal wind and zonal-mean zonal wind standard deviation in a subset of model runs using 5 km wavenumber-2 topography (K2H5). Zonal-mean zonal wind is illustrated with black and gray contours with an interval of 5 ms^{-1} for westerly and easterly winds, respectively. The standard deviation of zonal-mean zonal wind is illustrated with color shadings every 5 ms^{-1}. Model runs are shown in the same order as in Figure 1.

2.3. Stratospheric Vortex Statistics and Definition of SSW Events

To characterize the climatological strength of the stratospheric polar vortex in each model run, the maximum of the time-mean and zonal-mean zonal wind at 6.6 hPa (model level closest to 10 hPa) in the NH is used. This maximum is not necessarily at the same latitude for each model run (see Figure 2); it can shift southward or northward to accommodate meridional shifts in vortex location between simulations. To evaluate the temporal variability of the stratospheric polar vortex, the standard deviation of zonal-mean zonal wind is taken at the same pressure and latitude as the time-mean maximum.

SSW events are defined as a reversal of the zonal-mean zonal wind (Charlton & Polvani, 2007) affecting the stratospheric polar vortex. In each run, a transition from westerlies to easterlies is evaluated at 6.6 hPa at the latitude where the climatological maximum is located. When two SSW events are detected within 30 days of each other, the second event is discarded not to double count the same event.

3. Results

3.1. Upper Versus Lower-Stratospheric Equilibrium Temperatures

A detailed evaluation of zonal-mean zonal wind climatology and its temporal variability is first performed for the model runs forced with wavenumber-2 topography. A topography of 5 km is used in these runs (i.e., K2H5 runs) as it is found to promote a weaker mean state of the stratospheric vortex and large temporal variability, thus enhancing the likelihood of SSW events. A detailed analysis of the impact of topography on vortex variability and strength is shown later in section 3.3. Figure 2 illustrates the climatological structure of the zonal-mean zonal wind and its temporal variability in the troposphere and stratosphere. In most model runs, the stratospheric polar vortex is located at around 70°N which is slightly poleward from the location of the stratospheric polar vortex in observations (Jucker et al., 2013). This small discrepancy, however, should not be a serious concern for the idealized modeling study. Other model configurations used previously also suffer from small differences in vortex latitudes with respect to observations (compare Figures 1 and 2 of Jucker et al., 2013). In the troposphere, the jet typically shows a structure similar to observations with a 300 hPa

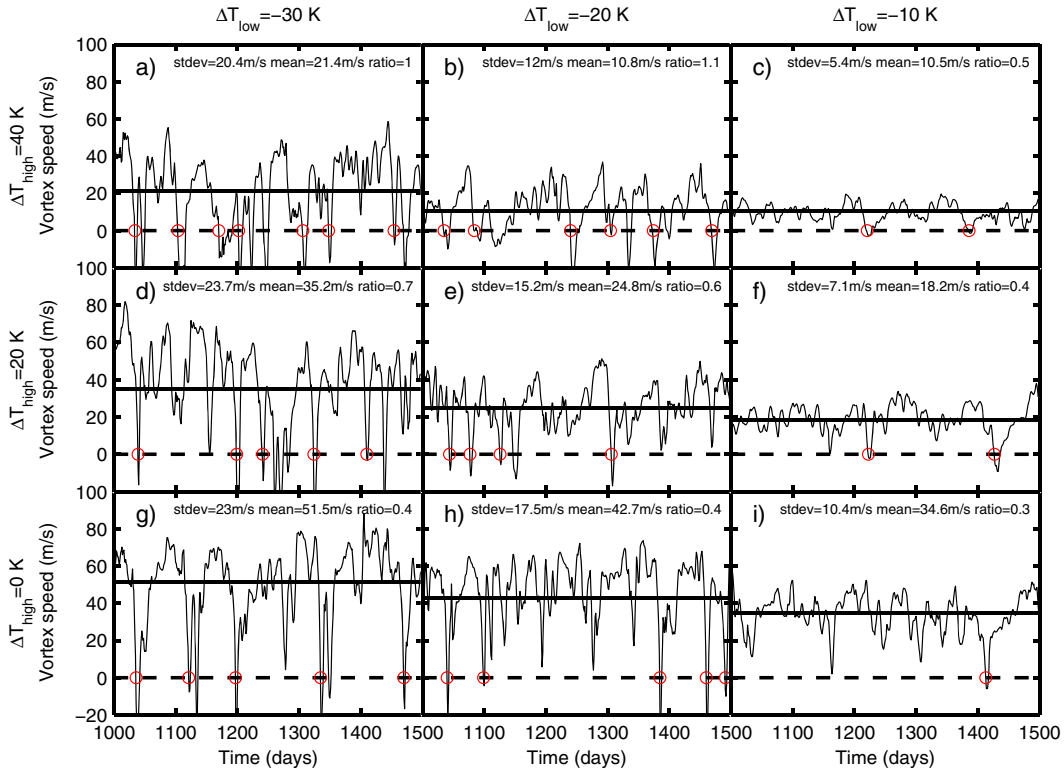

Figure 3. Time evolution of zonal-mean zonal wind at the latitude of the climatological maximum at 6.6 hPa for a subset of the 5 km wavenumber-2 model runs (K2H5). The wind is shown over a period of 500 days. The zero-wind line and the zonal-mean zonal wind climatology are shown using horizontal dashed and black solid lines, respectively. Sudden stratospheric warming events are indicated with red circles. The standard deviation of zonal wind, its time-mean, and their ratio are indicated in each panel. Model runs are shown in the same order as in Figure 1.

maximum located at around 30°N. A detailed inspection of zonal wind evolution in the upper troposphere (not shown here) reveals that the jet fluctuates around a single preferred location in the NH, without any hint of regime behavior as observed in Chan and Plumb (2009). Stratospheric zonal-mean zonal wind is maximized at around 3 hPa in the reference T_{eq} profile ($\Delta T_{low} = -20$ K and $\Delta T_{high} = 20$ K; Figure 2e) with a speed of 25 ms^{-1}. In most simulations, the maximum of zonal wind variability roughly coincides with the region of maximum winds with a small poleward shift.

A comparison of zonal-mean zonal wind across the simulations (Figure 2) reveals that the relaxation temperatures in the upper and lower polar stratosphere can have remarkable impact on the mean state and variability of the stratospheric polar vortex. First, looking at the effect of ΔT_{high} (comparing rows), it is noticeable that the strength of the stratospheric polar vortex diminishes as ΔT_{high} is increased (Gerber & Polvani, 2009; Polvani & Kushner, 2002). As the mean state of the vortex is weakened, it is also accompanied by a reduction of the variability. It is important to note, however, that the largest effect of ΔT_{high} on variability is found in the upper stratosphere. The lower to midstratospheric variability is largely unaltered. At first glance, lower-stratospheric relaxation temperature (ΔT_{low}; comparing columns) has a qualitatively similar impact on vortex variability although it has a slightly stronger impact on midstratospheric variability. When ΔT_{low} is warmer, the stratospheric vortex is weaker and less variable. Although not shown here, ΔT_{low} and ΔT_{high} have similar impacts on stratospheric polar vortex characteristics in model runs with wavenumber-1 topography (K1H5 runs), the main difference being an overall weaker variability of the stratospheric polar vortex.

It is not clear from Figure 2 how modifications of the T_{eq} affect the frequency of SSW events since a strengthening of the polar vortex and an increase of its variability are possibly two competing factors for SSW occurrence. The effect of stratospheric relaxation temperatures on the frequency of SSW events in K2H5 runs is more explicitly explored in Figure 3 which shows how ΔT_{low} and ΔT_{high} affect the evolution of zonal-mean zonal wind at the latitude of the climatological maximum at 6.6 hPa. For a fixed ΔT_{high} of 40 K (Figures 3a–3c), we see that a colder ΔT_{low} favors a stronger vortex that is subject to frequent and more intense reversals of the

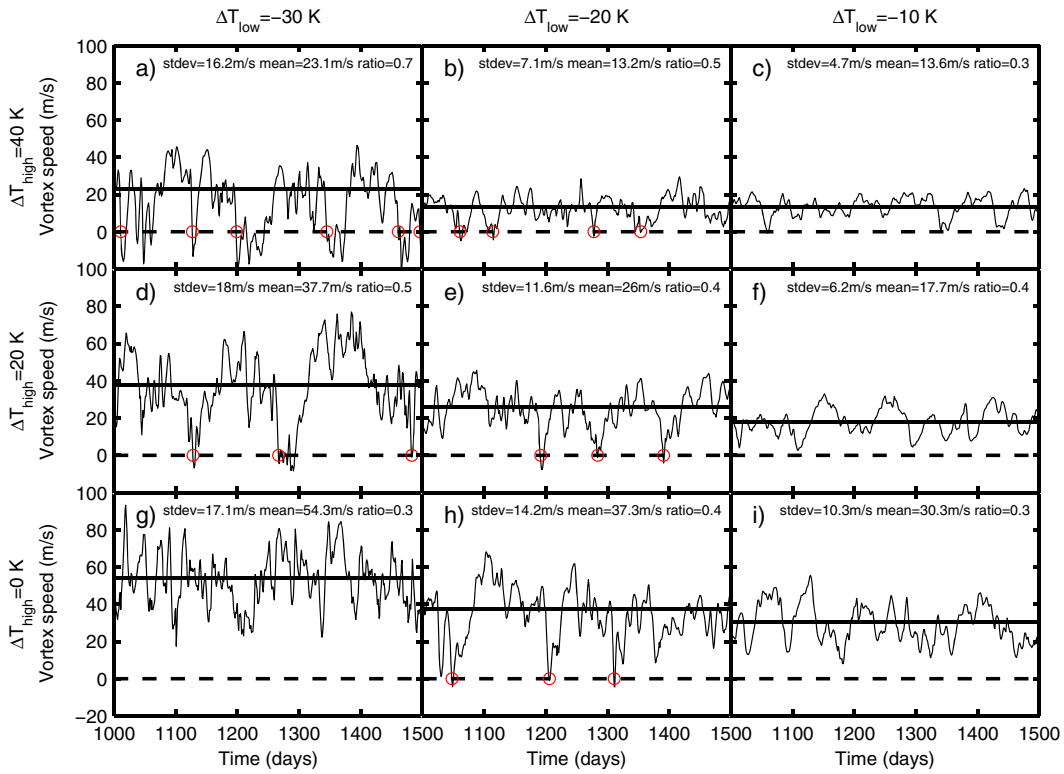

Figure 4. Same as Figure 3 but showing K1H5 runs.

zonal wind due to a larger increase in temporal variability as compared to the mean state. Decreasing ΔT_{high} while keeping ΔT_{low} constant (Figures 3b, 3e, and 3h for instance) also increases the strength of the vortex and its variability, but the effect on the frequency of SSWs is less obvious. Inspection of individual SSW events (not shown) reveals that almost all of them are vortex splits with rare displacements.

Time series of the strength of the stratospheric polar vortex in the runs forced by wavenumber-1 topography (K1H5 runs; Figure 4) are markedly different from those forced by wavenumber-2 topography (Figure 3). One notices that despite a similar time-mean vortex strength, SSW events are comparatively rare in K1H5 runs due to a substantially weaker vortex variability in comparison to K2H5 runs. Similar to K2H5 runs, vortex variability in these runs increases as ΔT_{low} is decreased but the effect of ΔT_{high} is less clear, especially when ΔT_{low} is set to −30 K. No SSW event occurs when ΔT_{low} and ΔT_{high} are at their coldest settings. Overall, more SSW events are observed when ΔT_{low} is cold and ΔT_{high} is warm due to the weaker basic-state wind and enhanced temporal variability. Unlike the runs forced with wavenumber-2 topography, wavenumber-1 topography runs favor the occurrence of SSW events characterized by a displacement of the vortex (not shown). This is in contrast to Sheshadri et al. (2015) who found that wavenumber-1 topography could generate both displacement and split events.

The effects of varying ΔT_{high} and ΔT_{low} on the vortex mean strength, variability, and the frequency of SSW events are further quantified in Figure 5 for both K1H5 and K2H5 simulations. It is clear from this figure that both colder ΔT_{high} and colder ΔT_{low} contribute to a strengthening of the polar vortex in the midstratosphere (Figures 5a and 5b). Both parameters also promote a larger temporal variability of the vortex at their coldest settings, but vortex variability is discernibly more sensitive to ΔT_{low} than ΔT_{high} as revealed by the contours of variability which are steeper in the ΔT_{low}-ΔT_{high} plane in comparison to the vortex strength contours (Figures 5c and 5d). One may thus expect that decreasing ΔT_{low} (larger variability) and increasing ΔT_{high} (weaker vortex) would be an optimal combination for the occurrence of SSW events. As expected, the frequency of SSW events is increased when the lower stratosphere (ΔT_{low}) is cold and the upper stratosphere (ΔT_{high}) is warm (Figures 5e and 5f). Similar sensitivities are found in both K1H5 and K2H5 runs, although SSW events are less frequent (about 50% less frequent) in the latter (comparing Figures 5a, 5c, and 5e to

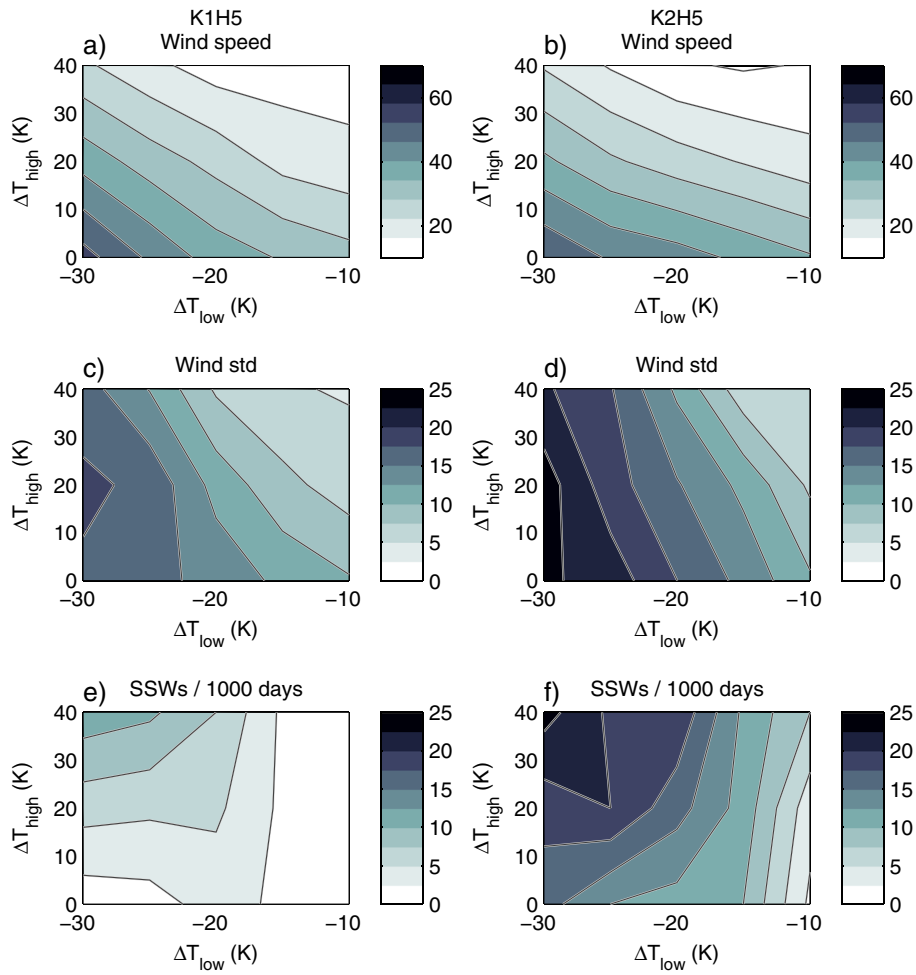

Figure 5. Response of (a, b) the zonal-mean zonal wind maximum at 6.6 hPa, (c, d) the zonal-mean zonal wind standard deviation at the same level and latitude, and (e, f) the frequency of sudden stratospheric warming (SSW) events (number of events occurring over a period of 1,000 days) to changes in the upper (ΔT_{high}) and lower (ΔT_{low}) stratospheric T_{eq} parameters. Statistics are reported only for the 5 km topography runs and are shown for (a, c, and e) wavenumber-1 and (b, d, and f) wavenumber-2 topographies. Zonal-mean zonal wind is contoured every 6 ms^{-1}. Zonal-mean zonal wind standard deviation is contoured every 2.5 $ms^{-}1$. The frequency of SSW events is contoured every 2.5 event per 1,000 days.

Figures 5b, 5d, and 5f). It is also important to note that a colder ΔT_{low} not only increases the frequency of SSW events but also makes them stronger (the deceleration of zonal-mean zonal wind during SSW events is larger; see Figures 3 and 4).

So far, the role played by the stratospheric mean state to regulate its variability was investigated by varying ΔT_{high} and ΔT_{low} independently. However, the control that these two parameters exert on the stratospheric circulation is not necessarily localized and could have a remote influence through processes responsible for maintaining a balanced flow and eddy feedbacks. Inspection of Figure 2 in fact reveals that varying ΔT_{low} alters the climatological zonal-mean zonal wind in both the lower and upper stratosphere. Likewise, small differences in lower-stratospheric winds are observed for different ΔT_{high} configurations. It is thus not completely clear which region of the stratosphere is most important to regulate stratospheric variability since changes in wave-mean flow interactions among the various model runs could result from either changes in the upper- or lower-stratospheric winds or both.

Figure 6 shows the correlation between zonal wind variability at 6.6 hPa and several climatological features at various pressure levels for K1H5 and K2H5 runs. It is found that the variability of the stratospheric polar vortex is strongly correlated with upward fluxes of wave activity, evaluated using EP flux (Edmon et al., 1980), and zonal-mean zonal wind in the lower stratosphere. On the other hand, it is strongly anticorrelated to polar cap

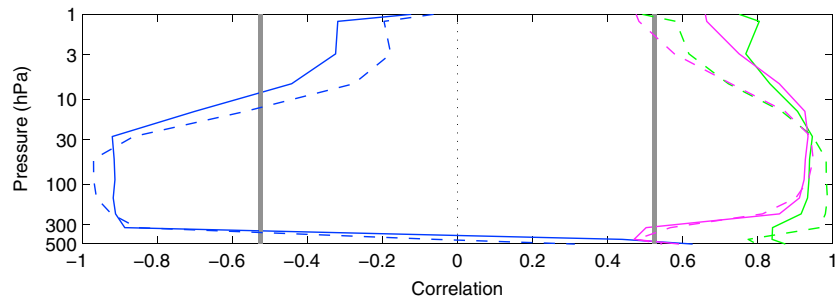

Figure 6. Correlation between the standard deviation of zonal-mean zonal wind at the latitude of its climatological maximum at 6.6 hPa and vertical EP flux averaged over 45°N–85°N (green, only the contributions of wavenumbers 1 and 2 are kept), zonal-mean zonal wind averaged over 45°N–85°N (magenta), and polar cap temperatures averaged over 70°N–90°N (blue) as a function of vertical level. Correlations are taken among model runs using 5 km topography for both wavenumber-1 (solid; K1H5) and wavenumber-2 (dashed; K2H5) topographies. Lower bounds for statistically significant correlations at a 95% confidence level are indicated with gray thick lines.

temperatures at the same level. For all quantities, the correlation is maximized at around 100 hPa and reduced in the troposphere and upper stratosphere. Similar correlations are also obtained with smaller topography amplitudes (not shown). This result indicates that the lower-stratospheric mean state is most strongly connected to the temporal variability of the polar vortex. Significant correlations with EP flux and zonal-mean zonal wind are also seen at higher altitudes indicating a connection, albeit weaker, with midstratospheric variability. It is not clear, however, how stronger zonal winds in the lower stratosphere can enhance upward fluxes of wave activity. Furthermore, an increase in time-mean wave activity flux alone may not explain a change in variability. The enhanced frequency of breakdowns and recoveries of the stratospheric polar vortex is likely achieved through more frequent and more intense episodes of planetary-wave propagation to the stratosphere (Limpasuvan et al., 2004; Martineau & Son, 2015; McDaniel & Black, 2005). It is conceivable that a strengthening of the basic-state zonal-mean zonal wind between the troposphere and the stratosphere may reduce the likelihood of wind reversal in that region during episodes upward planetary-scale wave propagation. By avoiding the formation of a critical layer resulting from zonal-mean wind reversal in the lower stratosphere, larger amounts of wave activity may propagate through the lower stratosphere and produce more frequent and stronger SSW events.

3.2. Eddy Forcing

In this section, the control exerted by the mean stratospheric state onto wave-mean flow interactions is investigated in more detail. Figure 7 shows cross sections of EP flux for various stratospheric T_{eq} parameters. The reference run shown in Figure 7, that is, K2H5 run with $\Delta T_{high} = 20$ K and $\Delta T_{low} = -20$ K, exhibits large upward EP flux in the upper troposphere (Figure 7e). Comparatively weaker upward EP flux is found in the stratosphere, with maximum upward fluxes located at around 50°N, slightly south of the stratospheric polar vortex (compare Figures 2e and 7e). The reduction of upward fluxes between the troposphere and stratosphere is concomitant to the large EP flux convergence observed at the tropopause, indicating that it is a natural barrier to vertical propagation of wave activity. Relatively large EP flux convergence is also observed in the midstratosphere, reflecting the time-mean deposition of easterly momentum due to wave breaking.

Investigating the effect of changing ΔT_{low} on the EP flux, it is evident that cooling the lower stratosphere enhances upward EP flux at all levels (comparing left and right columns of Figure 7). With a colder ΔT_{low}, the enhanced vertical flux is also accompanied with an enhanced convergence of EP flux in the midstratosphere, indicative of intensified planetary-scale wave breaking in the stratosphere. In comparison, altering ΔT_{high} has a subtle effect on EP flux with slightly enhanced upward fluxes in the center of the stratospheric polar vortex for a colder ΔT_{high} (comparing top and bottom rows). The sensitivity of EP flux divergence to ΔT_{high} is also much weaker compared to its sensitivity to ΔT_{low}.

Changes in the time-mean fluxes of wave activity and the drag they exert in the stratosphere do not, however, guarantee a change in stratospheric variability. In fact, it is well known that SSW events occur because of highly anomalous and short-lived bursts of planetary-scale wave activity propagating from the troposphere to the stratosphere (Limpasuvan et al., 2004; Martineau & Son, 2015; Polvani & Waugh, 2004) and that their propagation and interaction with the mean flow in the stratosphere are influenced by the stratospheric

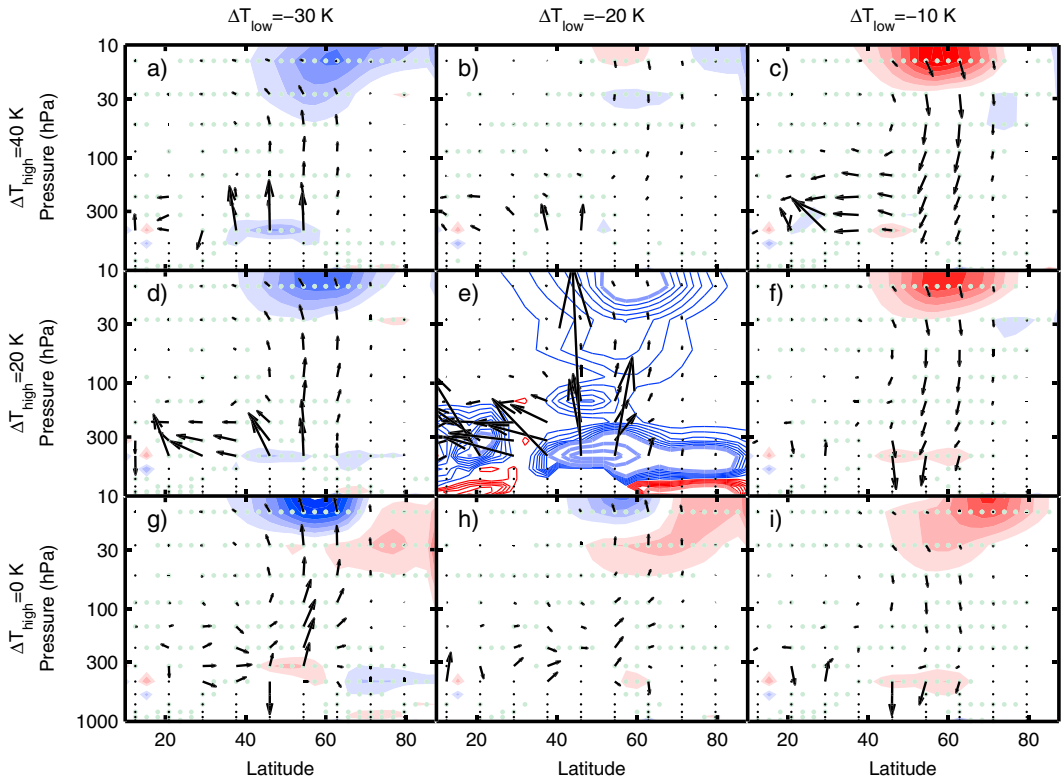

Figure 7. Latitude-pressure cross sections of EP flux are shown for the stratospheric T_{eq} configurations of Figure 1. Only K2H5 runs are shown. In panel (e), EP flux divergence is shown with color shading from −3.5 to 3.5 $ms^{-1}d^{-1}$ with 0.5 $ms^{-1}d^{-1}$ intervals. Blue and red contours are used for negative and positive quantities, respectively. Thicker contours are used for intervals larger than 4 $ms^{-1}d^{-1}$ with steps of 4 $ms^{-1}d^{-1}$. EP flux is shown with arrows using the scaling described in Edmon et al. (1980) and includes only the contributions from wavenumbers 1 and 2. The distance occupied by 10° latitude is equivalent to 2.3×10^{18} m^3 Pa, and the distance from 300 to 100 hPa is equivalent to 1.5×10^{19} m^3 Pa. Other panels show the differences with respect to the reference run in panel (e). EP flux arrows are magnified by a factor of 5 when differences from the reference are shown. Positive and negative differences in EP flux divergence are shown with color shading intervals of 0.5 $ms^{-1}d^{-1}$ in red and blue, respectively. Differences of EP fux divergence that are statistically significant at a 95% confidence interval according to a two-sample Student's *t* test are indicated with pale gray dots. Only EP flux differences whose meridional or vertical components are significant, determined with the same test as EP flux divergences, are shown.

state itself (Scott & Polvani, 2004; Scott & Polvani, 2006; Yoden et al., 2002). It is thus the frequency of these extremes, and not the time-mean forcing itself, that explains changes in polar vortex variability and the likelihood of SSW events. It is expected that the model runs that feature more frequent and intense SSW events present an enhanced probability of extreme upward EP flux associated with enhanced upward propagation of planetary-scale waves in the stratosphere. This relationship is verified in Figure 8 by illustrating the probability density function (PDF) of vertical fluxes of wave activity from the troposphere to the stratosphere (88.2 hPa). In addition to lower-stratospheric fluxes, the source of planetary-scale waves in the troposphere is investigated using the PDF of EP flux at 442.6 hPa. It is found that the PDF of EP flux at 442.6 hPa is largely insensitive to ΔT_{low} and ΔT_{high}. The generation of planetary-scale wave activity in the troposphere is therefore not a major factor explaining differences in the frequency of SSW events among the various model runs.

On the other hand, the distribution of vertical EP flux in the lower stratosphere changes markedly among the model runs. It is mostly affected by ΔT_{low} and is not sensitive to ΔT_{high}. For a warmer ΔT_{low}, the distribution of vertical wave activity flux is relatively narrow (Figures 8c, 8f, and 8i). In comparison, the PDF is markedly broader for a cold ΔT_{low} (Figures 8a, 8d, and 8g). The frequency of both positive and negative extremes is enhanced in comparison to the PDF of model runs with a warm lower stratosphere. This indicates that a cold lower-stratospheric mean state favors more frequent extremes of upward fluxes of wave activity in the stratosphere. Another interesting difference is the enhanced likelihood of downward wave activity fluxes (negative EP flux) in the lower stratosphere which could be linked to the downward reflection of

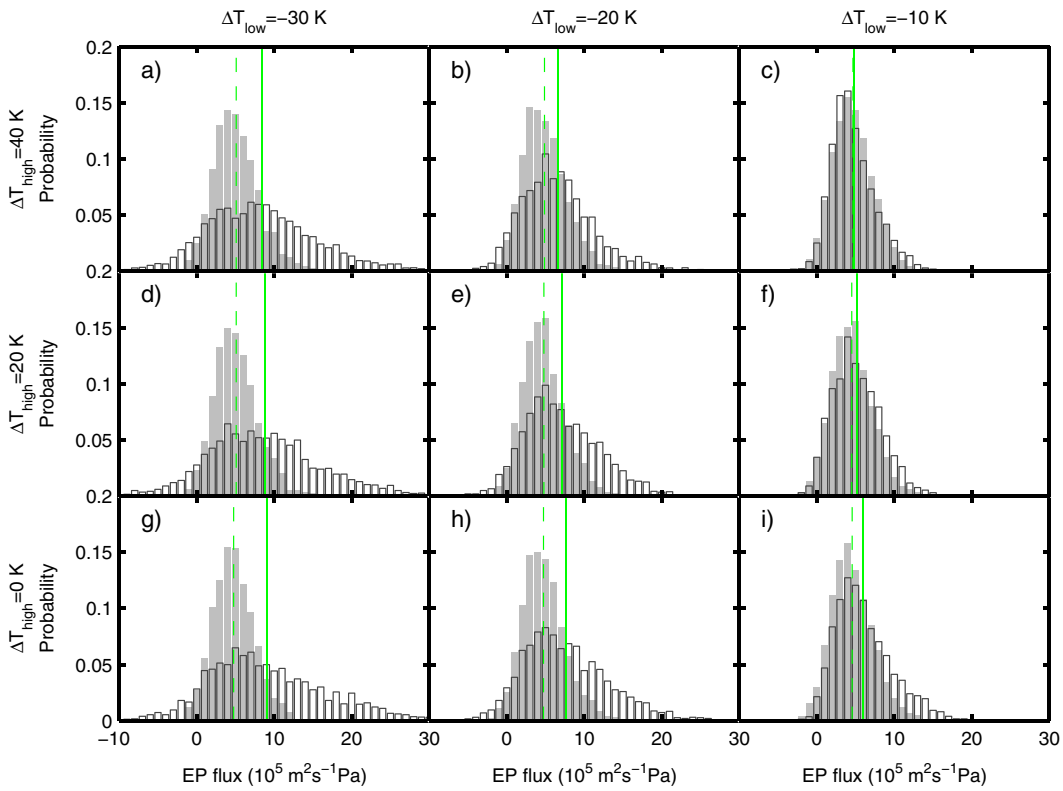

Figure 8. Probability density function of the vertical component of EP flux averaged from 45°N to 85°N at 442.6 hPa (shading) and 88.2 hPa (borders). Only K2H5 runs are shown. EP flux at 442.6 hPa is divided by 10 in order to fit the distributions at the two pressure levels on the same axis. Only the planetary-scale components (wavenumbers 1 and 2) of EP flux are considered. The average of the PDF of EP flux is shown with dashed (442.6 hPa) and solid (88.2 hPa) vertical lines. Model runs are shown in the same order as in Figure 1.

planetary-scale waves (Shaw & Perlwitz, 2013). It is, however, unclear how the lower-stratospheric equilibrium temperature is related to the frequency of negative wind shear events in the upper stratosphere, which was proposed to favor downward wave reflection (Shaw & Perlwitz, 2013). The fact that upward fluxes of wave activity are similar at the source altitude (442.6 hPa) but quite different in the lower stratosphere (88.2 hPa) for different ΔT_{low} suggests that the state of the atmosphere in between these two levels acts as a valve regulating the vertical propagation of planetary-scale waves from the troposphere to the stratosphere.

3.3. Impact of Topography

We now evaluate the impact of topography on stratospheric variability in our parameter sweep experiment. The sensitivity of the stratospheric polar vortex to topography height is considered separately for model runs with cold and warm relaxation temperatures in the lower stratosphere (ΔT_{low}) since previous sections have shown that this region has a strong influence on the permeability of the lower stratosphere to planetary-scale waves emanating from the troposphere.

Figure 9 summarizes all topography experiments for $\Delta T_{low} = -10$ K and $\Delta T_{high} = 20$ K (magenta lines) or $\Delta T_{low} = -30$ K and $\Delta T_{high} = 20$ K (blue lines). The mean strength (Figure 9a) and the variability (Figure 9b) of the stratospheric polar vortex are shown as a function of topography amplitude. As in the previous sections, the strength of the vortex is evaluated using the time-averaged zonal-mean zonal wind at 6.6 hPa at the latitude of maximum westerlies of the climatology of each model run. The temporal variability of the polar vortex is measured using the standard deviation of zonal wind at the same pressure and latitude. A weakening of the vortex is observed with increasing topography amplitude for both wavenumber-1 and wavenumber-2 topographies (Gerber & Polvani, 2009; Taguchi & Yoden, 2002; Taguchi et al., 2001). This relationship is, however, nonlinear. The strength of the stratospheric polar vortex is not very sensitive to topography when its amplitude is small but decreases markedly at intermediate topography amplitudes. It eventually saturates at large topographies, a feature also noted in Gerber and Polvani (2009). Similar result was also reported

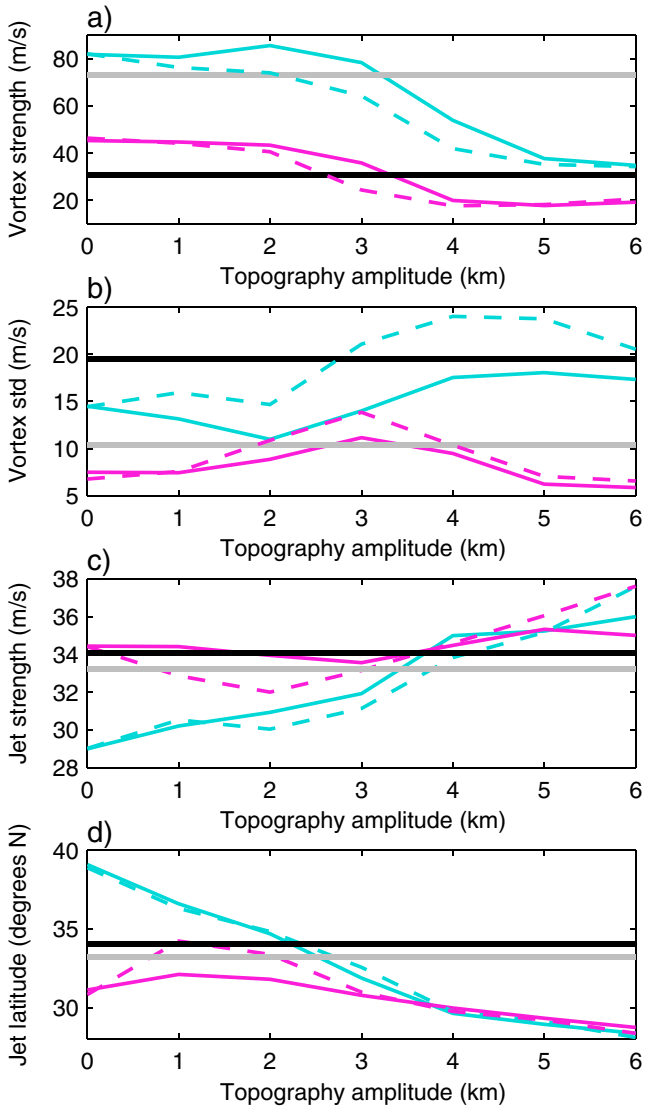

Figure 9. (a) The time-mean strength of the stratospheric polar vortex at 6.6 hPa and (b) its standard deviation are shown as a function of topography amplitude for lower-stratospheric equilibrium temperature configurations with $\Delta T_{low} = -10$ K (magenta) and $\Delta T_{low} = -30$ K (blue) for a ΔT_{high} fixed to 20 K. (c) The strength of the jet in the upper troposphere at 300 hPa and (d) its latitude are also reported. The relationships are shown separately for wavenumber-1 (solid lines) and wavenumber-2 (dashed lines) topographies. As a reference, NH and SH winter statistics from ERA-Interim are reported in black and gray, respectively.

for the tropospheric eddy kinetic energy (Son et al., 2009). The effect of topography is also sensitive to ΔT_{low}: the vortex weakens more strongly with enhanced topography when the lower-stratospheric relaxation temperatures are colder and also saturates at larger topography amplitudes. While for small topography amplitudes the strength of the vortex with $\Delta T_{low} = -30$ K is comparable to SH wintertime climatology, large topography amplitudes bring the vortex mean strength to a state more comparable to the NH wintertime vortex. In comparison, the strength of the vortex is much weaker when ΔT_{low} is fixed to -10 K and never reaches a strength similar to the SH basic state.

The amplitude of topography also affects strongly the temporal variability of the vortex. Vortex variability generally increases as the topography increases until it reaches a maximum and then subsequently decreases when topography is further increased (Figure 9b), a feature that was noted in previous modeling experiments (Gerber & Polvani, 2009). An exception is the initial decrease of variability between 0 and 2 km observed in the $\Delta T_{low} = -30$ K runs. Interestingly, lower-stratospheric relaxation temperatures have a large impact on the sensitivity of vortex variability with respect to topography amplitude. While the increase of variability with increased topography amplitude is small for $\Delta T_{low} = -10$ K and peaks at low topography amplitudes, changes in variability are large for $\Delta T_{low} = -30$ K and do not peak until a topography amplitude of 4–5 km. The variability is also markedly larger in wavenumber-2 runs in comparison to wavenumber-1 runs, a feature that is not conspicuous in $\Delta T_{low} = -10$ K runs. Unlike Gerber et al. (2009) who observed a transition from a strong undisturbed vortex to a vortex that is completely destroyed upon increasing wavenumber-1 topography amplitude, our model settings yield regimes with large variability (and frequent SSW events) even with large-amplitude wavenumber-1 topographies (see also Figure 11). In general, the influence of ΔT_{high} on the sensitivity of the stratospheric polar vortex variability to topography is weak compared to the effect of ΔT_{low} (not shown).

Because it affects directly the atmospheric circulation in the troposphere, topography has also an important impact on the climatological state of the jet. As mentioned in Gerber and Polvani (2009), the presence of topography can alter not only the momentum balance by imposing a direct drag on the winds but also the tropospheric eddies which also play an important role in regulating the momentum budget in the troposphere. Figures 9c and 9d show the sensitivity of the strength and latitude of the jet with respect to topography amplitude in our simulations. For larger topography amplitudes, the jet strengthens as it is also pulled equatorward (Gerber & Polvani, 2009). This response is not very sensitive to the topography wavenumber. A cold ΔT_{low} increases the sensitivity of the jet to the topography amplitude. The difference is observed mainly at small topography amplitudes where a colder lower stratosphere favors a weaker jet that is shifted to the north in comparison to a warmer ΔT_{low} configuration. This response, similar to the tropospheric response to ozone depletion-like stratospheric cooling of Sheshadri and Plumb (2016), is not a pure annular mode response since it involves both a shift of the jet and a modulation of its amplitude. At large topography amplitudes, however, the characteristics of the jet are similar regardless of the lower-stratospheric temperatures. Other than topography amplitude and zonal wavenumber, the meridional position of topography relative to the tropospheric jet may also have an important influence on both tropospheric and stratospheric circulations. This influence, which is not documented here, will be explored in future work.

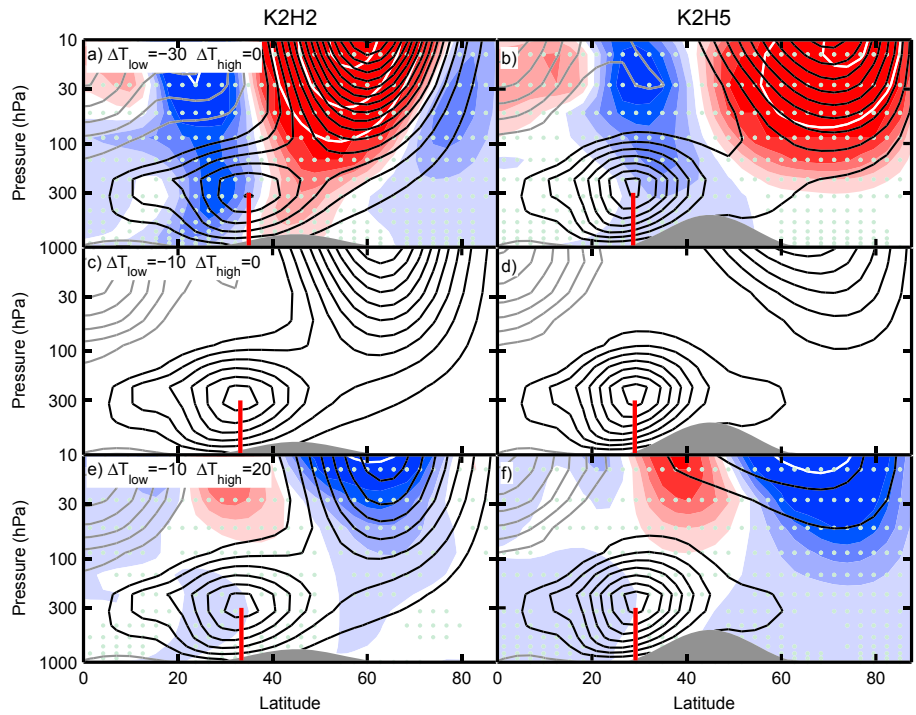

Figure 10. The sensitivity of zonal-mean zonal wind to ΔT_{low} and ΔT_{high} is illustrated for (a, c, e) 2 km topography runs (K2H2 runs) and (b, d, f) 5 km topography runs (K2H5 runs) for wavenumber-2 topography. For both topography amplitudes, panels (c) and (d) serve as a reference ($\Delta T_{low} = -10$ K and $\Delta T_{high} = 0$ K). ΔT_{low} is changed to −30 K in (a) and (b), and ΔT_{high} is changed to 20 K in (e) and (f). Zonal-mean zonal wind is contoured every 5 ms^{-1} using gray and black contours for easterly and westerly winds, respectively. The difference with respect to the reference runs is illustrated with color shadings from −5 to 5 every 1 ms^{-1} using blue and red for negative and positive values, respectively. Larger differences are shown with white lines every 10 ms^{-1}. The latitude of the jet at 300 hPa is reported with vertical red lines. Differences from the reference runs that are significant at a 95% confidence level using a two-sample Student's *t* test are indicated with pale gray dots.

3.4. Vertical Coupling

This section briefly examines the impact of the different parameters of the parameter sweep experiment on the vertical coupling between the troposphere and the stratosphere. Figure 10 shows the pressure-latitude structure of the zonal-mean zonal wind for various T_{eq} configurations in the stratosphere (rows) and two topography amplitudes (columns). The figure shows wavenumber-2 topography runs, but the results are similar for wavenumber-1 topography (not shown). In 2 km topography runs (K2H2 runs), the zonal-mean zonal wind response to a cooling of the lower stratosphere (Figure 10a) shows a horizontal tripole pattern spanning both the stratosphere and troposphere. Similar to the annular mode signature (Thompson & Wallace, 2000), this pattern represents a strengthening of the stratospheric polar vortex and a poleward shift of the tropospheric jet. This response is in agreement with model simulations where ozone depletion-like cooling is introduced in the lower stratosphere (Yang et al., 2015). In addition to the poleward shift, the time-mean strength of the jet is also weakened (see also Figures 9c and 9d). Although modifying ΔT_{low} does not alter directly the temperature profile in the vicinity of the jet, it may have an indirect influence through eddy feedback (Chen & Held, 2007; Kushner & Polvani, 2004) or through the maintenance of balance following a rearrangement of stratospheric potential vorticity (PV) by eddies (e.g., Black & McDaniel, 2004; McDaniel & Black, 2005).

In comparison to cooling the lower stratosphere, warming the upper stratosphere (Figure 10e) has a very little impact on the tropospheric jet. Only a weak deceleration is observed to the south of the jet. The changes in zonal-mean zonal wind are mostly confined to the stratosphere where the vortex is decelerated. Although of weaker amplitude, the response pattern in the stratosphere shows a structure that is similar (but of opposite sign) to the lower-stratospheric cooling response.

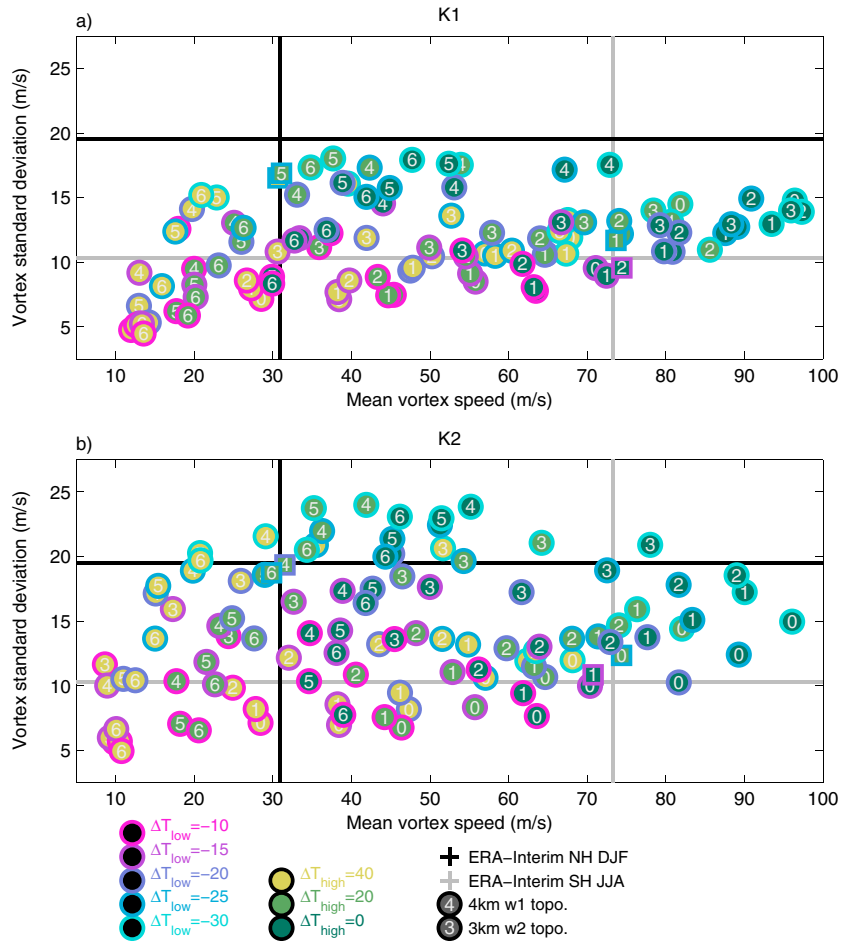

Figure 11. The strength of the stratospheric polar vortex and its variability at 6.6 hPa are indicated for all model runs of the parameter sweep experiment. Each model run is labeled according to the legend. Stratospheric polar vortex characteristics from ERA-Interim during NH winter and SH winter are reported with black and gray lines, respectively. Wavenumber-1 and wavenumber-2 runs are reported in panels (a) and (b), respectively. Several runs that are further compared with reanalysis data in Figure 12 are shown with squares.

When repeating the experiment with larger topography (5 km; Figures 10b, 10d, and 10f), the sensitivity of the tropospheric jet to lower-stratospheric temperatures is greatly reduced. The response is actually opposite to the 2 km topography response with deceleration located to the north of the jet. The nature of the response is also altered in the stratosphere. The different response likely reflects changes in the vortex climatology between 2 km and 5 km topography runs. The response to upper-stratospheric warming also changes between 2 km and 5 km runs. The weak deceleration to the south of the jet in the 2 km run (Figure 10e) becomes an almost hemispheric-wide deceleration of tropospheric winds in the 5 km run (Figure 10f). Zonal asymmetries were found to reduce the time scale of tropospheric jet variability in a similar model (Gerber & Vallis, 2007). This could, according to the fluctuation-dissipation theorem, reduce the sensitivity of the circulation to external forcings. Gerber and Polvani (2009) have observed, as expected, a reduction in the sensitivity of the tropospheric jet to stratospheric vortex strength with the introduction of topography.

3.5. A Comparison With Observations

This section evaluates how the time-mean strength and variability of the stratospheric polar vortex in each run of the parameter sweep experiment compare to the wintertime state of the NH and SH stratospheric vortices in observations. In ERA-Interim, boreal winter statistics are based on December-January-February data and austral winter is based on June-July-August. Figure 11 illustrates the strength and variability of the stratospheric polar vortex for each run of the parameter sweep experiment.

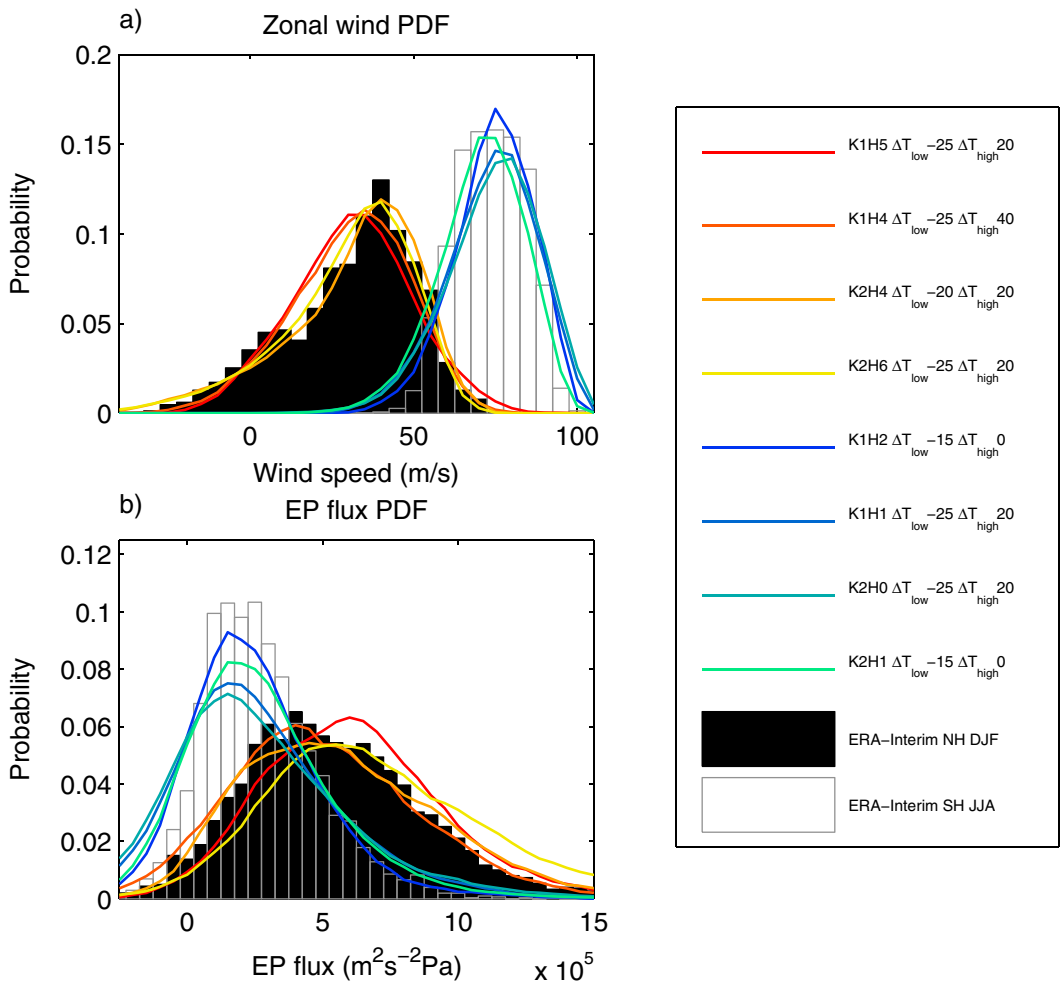

Figure 12. Probability density function of (a) zonal-mean zonal wind and (b) vertical component of EP flux for a selection of runs that have similar basic state and variability of the stratospheric polar vortex compared to observations in the NH and SH. EP flux is averaged from 20°N to 85°N at 88.2 hPa. Its sign is reversed in order to represent upward EP flux with positive sign. Zonal-mean zonal wind is sampled at the latitude of maximum climatological wind at 6.6 hPa. Model runs and reanalysis data are illustrated as shown in the legend. While model runs mimicking NH conditions are shown with warm colors, those mimicking SH conditions are shown with cold colors.

Overall, wavenumber-1 runs are less variable than wavenumber-2 runs and exhibit a smaller range of variability (Figures 11a and 11b; see also Figures 5c, 5d and 9b). It is also apparent that small topography runs (0–3 km) are typically characterized by a strong vortex, whereas most of larger topography runs (4–6 km) are characterized by a weaker vortex (see also Figure 9a). The experiments that are closest to NH wintertime observations (intersection of black lines) are generally observed for a wide range of topography amplitudes (4–6 km) in both wavenumber-1 and wavenumber-2 runs. These runs have a point in common: a cold lower-stratospheric relaxation temperature. On the other hand, stratospheric vortex characteristics similar to SH wintertime conditions (intersection of gray lines) are obtained with small topography amplitudes (0–3 km). Wavenumber-1 runs are closer to SH observations in comparison wavenumber-2 runs because of a generally less variable polar vortex. NH-like and SH-like conditions are likely obtainable with both wavenumber-1 and wavenumber-2 topographies by extending the parameter sweep experiment. Further cooling lower-stratospheric relaxation temperatures would promote greater vortex variability in wavenumber-1 runs, reproducing better NH-like conditions.

The model runs that are most similar to NH and SH stratospheric polar vortex mean state and variability are further investigated in terms of the probability distribution of zonal-mean zonal wind and fluxes of wave activity between the troposphere and stratosphere (Figure 12). For illustrative purpose, two representative

runs are selected for both wavenumber-1 and wavenumber-2 topographies that have the smallest Euclidean distance with respect to SH and NH wintertime conditions in the vortex mean and vortex variability space (indicated with squares in Figure 11).

The runs that are selected to best represent NH winter climatology are characterized by a PDF of zonal-mean zonal wind that is strikingly similar to observations (Figure 12a). This distribution is maximized at around 40 ms^{-1} and is negatively skewed. Model runs that reproduce better SH mean vortex speed and variability also show a distribution of zonal wind that is similar to observations but peaks at about 75 ms^{-1}. Fluxes of wave activity from the troposphere to the stratosphere also show noticeable agreement between observations and model runs that capture NH polar stratospheric variability the best (Figure 12b). However, those that mimic SH variability show a slightly broader distribution in comparison to observations.

Despite the fact that the climatology and variability of the stratospheric circulation are not identical between the model runs and observations, the parameter sweep experiment proves to be useful to identify model configurations that closely mimic stratospheric polar vortex variability in observations, especially in light of the fact that many previous idealized modeling studies could not produce sufficiently frequent SSW events with wavenumber-1 topography (Gerber & Polvani, 2009; Sheshadri et al., 2015). The model settings that favor frequent SSW events and NH-like stratospheric vortex regimes with wavenumber-1 topography use relatively cold lower-stratospheric equilibrium temperatures ($\Delta T_{low} = -30$ K) and large topography amplitudes (4–6 km). The runs that best reproduce observations have a greater chance of capturing the wave-mean flow interactions occurring in the real atmosphere. The dynamics of the stratosphere-troposphere coupling on subseasonal time scale will be investigated further in future work.

4. Discussion

The sensitivity of stratospheric variability to the lower-stratospheric basic state observed in this study could partly explain why some previous modeling studies did not obtain frequent SSW events with wavenumber-1 topography (Gerber & Polvani, 2009; Sheshadri et al., 2015). Many recent studies using a model configuration similar to Held and Suarez (1994) in the troposphere have included equilibrium relaxation temperature settings representing the stratosphere only above 100 hPa (Gerber & Polvani, 2009; Jucker et al., 2014; Polvani & Kushner, 2002) or 200 hPa (Sheshadri et al., 2015), possibly missing out important features of the lower-stratospheric circulation. Transitioning from the tropospheric profile to the stratospheric profile at 200 hPa instead of 100 hPa may have been a favorable factor for the occurrence of SSW events when using wavenumber-1 topography in Sheshadri et al. (2015), although SSW events were still relatively infrequent in comparison to observations. Another likely factor is the introduction of a seasonal cycle, which resets the stratospheric circulation every winter and may prevent a perpetual breakdown of the vortex as observed by Gerber and Polvani (2009). Our results strongly suggest that the cold radiative equilibrium region in the polar lower stratosphere identified in Jucker et al. (2013) using radiative transfer calculations is an important feature that is required for a realistic variability of the stratospheric polar vortex in idealized numerical simulations.

While the parameter sweep experiment carried in this study has reproduced a diversity of vortex regimes by modifying constant parameters of the model, some previous studies have instead introduced a seasonal cycle of stratospheric temperatures (Kushner & Polvani, 2006; Sheshadri et al., 2015; Sun et al., 2011; Taguchi & Yoden, 2002). Not surprisingly, these studies highlight a sensitivity of stratospheric variability and vertical coupling to the seasonal cycle of temperatures in the stratosphere which produces distinct wave-mean flow interactions depending on the season. These studies have also noted a sensitivity to topographic forcing which can produce NH- or SH-like conditions by modulating fluxes of planetary-scale wave activity between the troposphere and the stratosphere. Given the importance of the lower stratosphere highlighted in this work, it seems worthwhile that future studies consider more carefully the seasonal cycle in the lower stratosphere.

In terms of the dynamical mechanisms explaining the control of stratospheric variability by the lower-stratospheric circulation, our results suggest that the lower stratosphere does not only act to permit or inhibit fluxes of planetary-scale wave activity from the troposphere to the stratosphere but also works, like a valve, to control the amount of wave activity able to propagate through. Refractive-index arguments, based on linear quasigeostrophic theory, were also investigated in this work (not shown) but could not successfully explain the changes in wave activity fluxes with respect to changes in the basic state. Our results suggest that the answer may not be found in the time-mean state of the circulation but rather in its variability: fluxes of wave

activity from the troposphere to the stratosphere can vary a lot as evidenced by the probability distribution of vertical EP flux. The relatively weak winds and quasigeostrophic PV gradients found in between the tropospheric jet and the stratospheric polar vortex could limit the strength of the bursts of planetary-scale wave activity that propagate through the lower stratosphere and disturb the stratospheric vortex (Martineau & Son, 2015). According to finite amplitude wave activity theory (Nakamura & Solomon, 2011), the sum of zonal wind and wave activity is constant if the residual circulation is of secondary importance (Solomon, 2014). This thus suggests that for a stronger zonal wind in the lower stratosphere, a wave packet of stronger amplitude may propagate through the region without reversing the zonal-mean zonal wind. This would allow larger amounts of wave activity to reach the stratosphere before a shutdown of the lower-stratospheric valve. This mechanism potentially explains why a weaker stratospheric vortex is not necessarily more easily reversed in our modeling experiment. Zonal wind reversal in the midstratosphere will not occur if the lower-stratospheric mean flow does not allow the propagation of wave packets of sufficiently large amplitude to exert the necessary wave drag in the middle and upper stratosphere. This hypothesis will be investigated in future work.

This control exerted by the lower stratosphere onto stratospheric variability may have implications for climate change. Temperature trends related to global warming show cooling in the polar lower stratosphere. In agreement with the sensitivities found in this work, this could explain, at least partly, the projected increase of the frequency of SSW events in a warm climate (Kim et al., 2017). To explain changes in the variability of the stratospheric polar vortex in CO2-doubling model experiments, Schimanke et al. (2013) proposed that stronger lower-stratospheric winds favor enhanced upward propagation of wave activity and greater stratospheric variability, in agreement with our results. Our results also suggest that a misrepresentation of the lower-stratospheric basic state could be a source of bias of stratospheric variability in reanalysis data, especially in the data sets that do not assimilate satellite observations such as National Oceanic and Atmospheric Administration's 20th Century Reanalysis, the Japanese 55-year Reanalysis assimilating conventional observations, and ECMWF's Atmospheric Reanalysis of the 20th Century. The role played by time-mean biases affecting upper-tropospheric and lower-stratospheric circulations in regulating variability biases is worthy of further investigations.

5. Conclusions

The sensitivity of the variability of the stratospheric polar vortex to lower- and upper-stratospheric mean states is investigated by carrying a parameter sweep experiment with a dynamical core GCM. The parameter space explored includes topographic amplitude and wavenumber and relaxation temperatures in the lower stratosphere and upper stratosphere. The stratospheric temperature profile is altered in the polar region, which acts to modify the meridional temperature gradient in the midlatitudes, which ultimately alters the strength of the stratospheric polar vortex through maintenance of thermal wind balance and eddy feedback.

The key result that emerges from our analysis is that the variability of the stratospheric polar vortex in the midstratosphere (6.6 hPa) depends critically on the temperature in the lower stratosphere. This result is in agreement with previous studies that have suggested that the upper troposphere and lower stratosphere regions act as a valve for the upward propagation of planetary-scale waves from the troposphere to the stratosphere (Chen & Robinson, 1992; Hitchcock & Shepherd, 2013; Scott & Polvani, 2006) and with recent modeling studies suggesting that the evolution of SSW events is sensitive to stratospheric conditions (de la Cámara et al., 2017; Hitchcock & Haynes, 2016). In our experiments, a cold polar lower stratosphere is key to obtain a large variability of the stratospheric polar vortex, which enhances the frequency of SSW events in model runs forced with both wavenumber-1 and wavenumber-2 topographies. SSWs, however, are about twice as frequent with wavenumber-2 topography. Lower-stratospheric temperatures also have an impact on the mean strength of the stratospheric polar vortex: a colder lower stratosphere strengthens the vortex. Although the stratospheric polar vortex strengthens when the lower-stratospheric basic state is cooled at the pole, SSWs become more frequent and more intense because the increased variability overpowers the increased strength.

On the other hand, upper-stratospheric temperatures were found to primarily control the mean strength of the stratospheric polar vortex with a relatively weak impact on vortex variability in comparison to lower-stratospheric temperatures. As in previous modeling studies, warming the upper stratosphere over the polar cap results in a weaker vortex (Gerber & Polvani, 2009; Kushner & Polvani, 2004; Polvani & Kushner, 2002). However, warming the upper stratosphere alone is not effective to produce more frequent SSW

events since it leads to not only a decrease of the mean vortex strength but also a decrease of variability (Gerber & Polvani, 2009).

Concerning possible dynamical mechanisms explaining the relationship between lower-stratospheric temperatures and midstratospheric variability, model runs with a colder polar cap in the lower stratosphere present a much broader spectrum of wave activity fluxes from the troposphere to the stratosphere, with a marked increase in extreme values of vertical fluxes of planetary-scale wave activity, an essential feature to cause SSW events (e.g., Limpasuvan et al., 2004; Martineau & Son, 2015; Polvani & Waugh, 2004). On the other hand, the probability distribution of vertical fluxes of planetary-scale waves in the upper troposphere is largely insensitive to the stratospheric equilibrium temperature. This suggests that some feature of the circulation in the lower stratosphere or upper troposphere must control the amount of wave activity allowed to radiate from the troposphere to the stratosphere.

Concerning the realism of these simulations in terms of stratospheric mean state and variability, we found that it is possible to obtain SH-like or NH-like conditions with model runs forced with both wavenumber-1 and wavenumber-2 topographies. While NH wintertime polar vortex conditions were obtained for large topography amplitudes and a generally cold lower stratosphere, SH wintertime polar vortex conditions were obtained with small topography amplitudes.

Further studies are necessary to better understand the detailed dynamical mechanism(s) responsible for the sensitivity of stratospheric variability to the lower-stratospheric basic state. The impact of lower-stratospheric conditions onto the two-way coupling between the troposphere and the stratosphere during SSW events will be the topic of a follow-up study. For instance, the role played by synoptic and planetary-scales waves in the tropospheric response to SSW events and their sensitivity to lower-stratospheric conditions could be further investigated using the methodology developed in Yang et al. (2015). The sensitivity of stratosphere-troposphere coupling to upper-tropospheric conditions also deserves more attention.

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